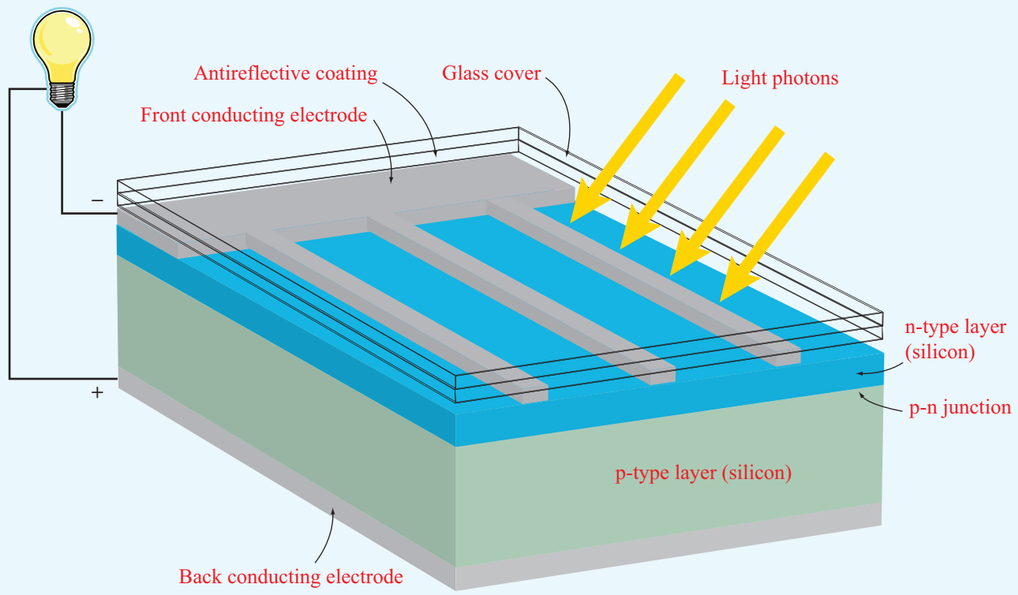


Figure TF2-2 Basic structure of a photovoltaic cell.

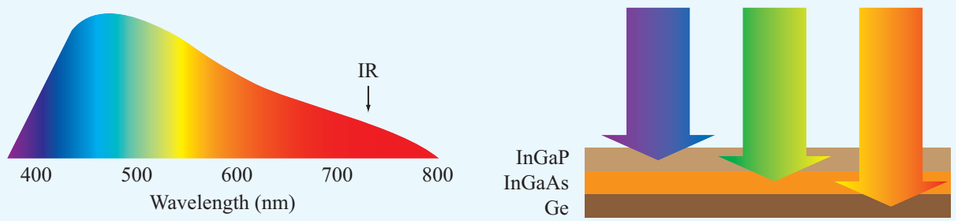
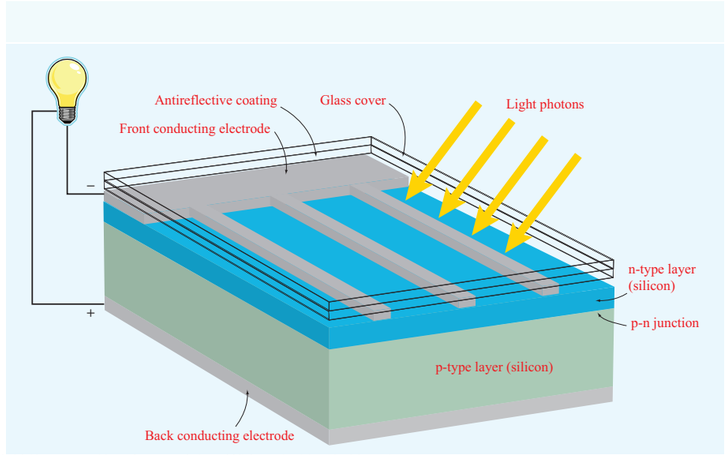


Figure TF2-3 In a multijunction PV device, different layers absorb different parts of the light spectrum.

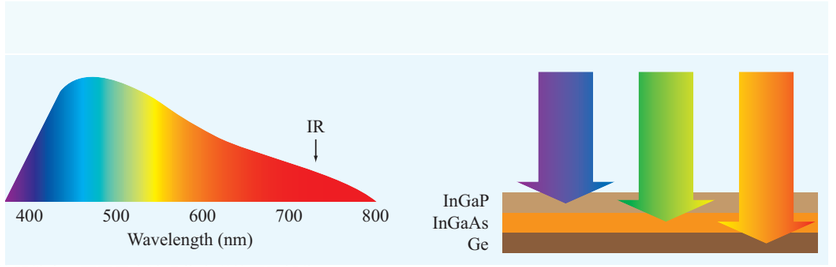
太阳能电池结构：图示与中文图例



图TF2-2 光伏电池的基本结构。原图的正负号、电极引线与层间关系保持不变。

图TF2-2 英中对照

英文标注	中文含义
Light photons	入射光子
Glass cover / Antireflective coating	玻璃盖板 / 减反射膜
Front / Back conducting electrode	前导电电极 / 背导电电极
n-type layer (silicon)	n型硅层
p-type layer (silicon)	p型硅层
p-n junction	p–n结



图TF2-3 多结光伏器件中，不同子电池吸收光谱的不同部分。

Wavelength（nm）：波长（纳米）；IR：红外。InGaP为磷化铟镓，InGaAs为砷化铟镓，Ge为锗。材料化学式中的In代表铟，不能把大写I误认作数字1或小写l。图中从短波到长波对应光子能量由高到低。

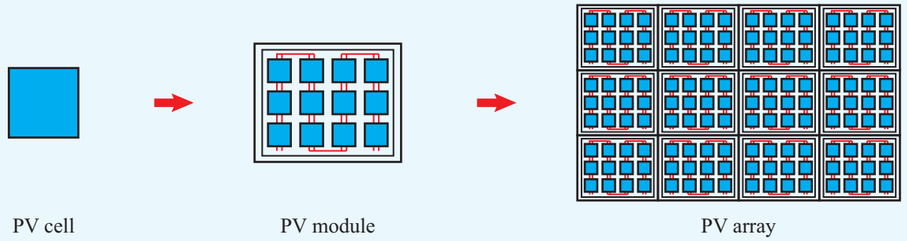


Figure TF2-4 PV cells, modules, and arrays.

► The multijunction technique offers an improvement in conversion efficiency of 2–4 times over that of the single-junction cell. However, the fabrication cost is significantly greater. ◄

Modules, Arrays, and Systems

A ***photovoltaic module*** consists of multiple PV cells connected together in order to supply electrical power at a specified voltage level, such as 12 or 24 V. The combination of multiple modules generates a ***PV array*** (**Fig. TF2-4**). The amount of generated power depends on the intensity of the intercepted sunlight, the total area of the module or array, and the conversion efficiencies of the individual cells. If the PV energy source is to serve multiple functions, it is integrated into an energy management system that includes a dc to ac current converter and batteries to store energy for later use (**Fig. TF2-5**).

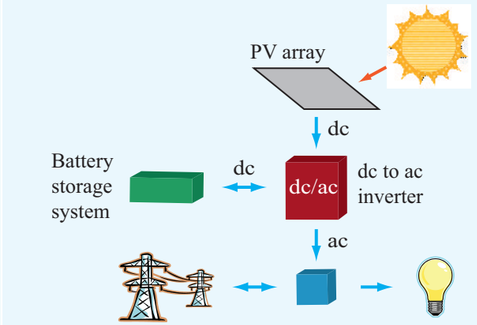
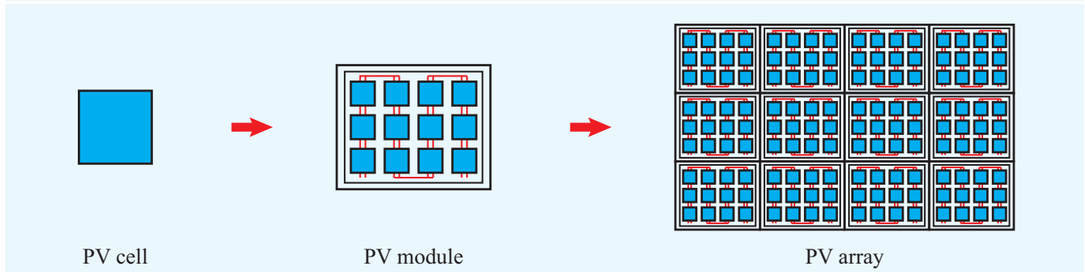


Figure TF2-5 Components of a large-scale photovoltaic system.

光伏组件、阵列与系统



图TF2-4 光伏电池（PV cell）、光伏组件（PV module）与光伏阵列（PV array）。

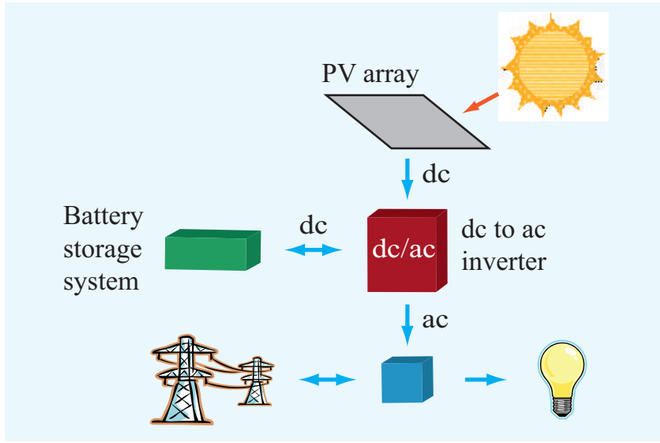
原书指出，多结技术可使转换效率达到单结电池的约2—4倍，但制造成本也显著增加。

比较口径校注：“达到约2—4倍”不等于“提高了2—4倍”。该数值是原书的概括性比较，未给出材料、光照和聚光条件，不能视为所有多结与单结器件间恒定的效率比例。

组件、阵列与系统

光伏组件由多个电池单元互连而成，以便在指定电压（例如12 V或24 V）下供电。多个组件组合成光伏阵列（图TF2-4）。其输出电功率取决于接收到的太阳辐照强度、组件或阵列总面积，以及各电池的转换效率。这里讨论的是功率，不能与一段时间内累计产生的电能混淆。

若光伏电源需要承担多种供能功能，可将其集成到能源管理系统中。图TF2-5所示系统包括把直流转换为交流的逆变器，以及将电能储存起来供以后使用的蓄电池系统。



图TF2-5 大型光伏系统的组成。

图例：PV array为光伏阵列；dc to ac inverter为直流–交流逆变器；Battery storage system为蓄电池储能系统；dc、ac分别表示直流、交流。图示是一种含储能的系统配置，并不表示所有光伏系统都必须配置蓄电池。

Exercise 1-9: Express the following complex functions in polar form:

$$z_1 = (4 - j3)^2,$$
$$z_2 = (4 - j3)^{1/2}.$$

Answer: $z_1 = 25\angle -73.7°$, $z_2 = \pm\sqrt{5}\angle -18.4°$. (See EM.)

Exercise 1-10: Show that $\sqrt{2j} = \pm(1 + j)$.

Answer: (See EM.)

1-7 Review of Phasors

Phasor analysis is a useful mathematical tool for solving problems involving linear systems in which the excitation is a periodic time function. Many engineering problems are cast in the form of linear integro-differential equations. If the excitation, more commonly known as the ***forcing function***, varies sinusoidally with time, the use of phasor notation to represent time-dependent variables allows us to convert a linear integro-differential equation into a linear equation with no sinusoidal functions, thereby simplifying the method of solution. After solving for the desired variable, such as the voltage or current in a circuit, conversion from the phasor domain back to the time domain provides the desired result.

The phasor technique also can be used to analyze linear systems when the forcing function is a (nonsinusoidal) periodic time function, such as a square wave or a sequence of pulses. By expanding the forcing function into a Fourier series of sinusoidal components, we can solve for the desired variable using phasor analysis for each Fourier component of the forcing function separately. According to the principle of superposition, the sum of the solutions due to all of the Fourier components gives the same result as one would obtain had the problem been solved entirely in the time domain without the aid of Fourier representation. The obvious advantage of the phasor–Fourier approach is simplicity. Moreover, in the case of nonperiodic source functions, such as a single pulse, the functions can be expressed as Fourier integrals, and a similar application of the principle of superposition can be used as well.

The simple RC circuit shown in **Fig. 1-20** contains a sinusoidally time-varying voltage source given by

$$v_s(t) = V_0 \sin(\omega t + \phi_0), \tag{1.55}$$

where V_0 is the amplitude, ω is the angular frequency, and ϕ_0 is a reference phase. Application of Kirchhoff's voltage law

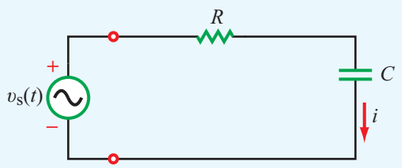


Figure 1-20 RC circuit connected to a voltage source $v_s(t)$.

gives the following loop equation:

$$R\,i(t) + \frac{1}{C}\int i(t)\,dt = v_s(t). \tag{1.56}$$

(time domain)

Our objective is to obtain an expression for the current $i(t)$. We can do this by solving Eq. (1.56) in the time domain, which is somewhat cumbersome because the forcing function $v_s(t)$ is a sinusoid. Alternatively, we can take advantage of the phasor-domain solution technique as follows.

1-7.1 Solution Procedure

Step 1: **Adopt a cosine reference**

To establish a phase reference for all time-varying currents and voltages in the circuit, the forcing function is expressed as a cosine (if not already in that form). In the present example,

$$\begin{aligned} v_s(t) &= V_0 \sin(\omega t + \phi_0) \\ &= V_0 \cos\left(\frac{\pi}{2} - \omega t - \phi_0\right) \\ &= V_0 \cos\left(\omega t + \phi_0 - \frac{\pi}{2}\right), \end{aligned} \tag{1.57}$$

where we used the properties $\sin x = \cos(\pi/2 - x)$ and $\cos(-x) = \cos x$.

Step 2: **Express time-dependent variables as phasors**

Any cosinusoidally time-varying function $z(t)$ can be expressed as

$$z(t) = \mathfrak{Re}\left[\widetilde{Z}\, e^{j\omega t}\right], \tag{1.58}$$

where $\widetilde{Z}$ is a time-independent function called the ***phasor*** of the ***instantaneous*** function $z(t)$. To distinguish instantaneous quantities from their phasor counterparts, a tilde ($\sim$) is added

1-7 相量复习

练习1-9： 将$z_1 = (4 - j3)^2$及$z_2 = (4 - j3)^{1/2}$写成极坐标形式。
答案： $z_1 = 25\angle(-73.7°)$；z_2的两根为$\pm\sqrt{5}\angle(-18.4°)$。详见配套电磁学资源（EM）。
练习1-10： 证明2j的两个平方根为$\pm(1 + j)$。核算：$(\pm(1 + j))^2 = 2j$；配套资源提供解答。

相量分析是处理线性系统正弦激励问题的数学工具。许多工程问题可写成线性积分微分方程；用相量表示随时间正弦变化的变量，可消去显式的正弦时间因子，将求解过程简化。求得电压、电流等量的相量后，再变换回时域。

对方波、脉冲序列等非正弦周期激励，可先作傅里叶级数展开，对每个正弦频率分量分别进行相量分析，再由叠加原理合成时域响应。这样得到的受迫响应与直接在时域求解一致，而步骤通常更简单。对单个脉冲等非周期激励，可采用傅里叶积分并叠加各频率分量。

适用条件校注：这里采用线性时不变系统及相应频率表示存在的条件；单一相量描述给定频率的正弦稳态响应，不能单独给出初始条件引起的暂态响应。不同频率须分别计算，不能把它们作为同频相量直接相加。

图1-20的串联RC电路由正弦电压源驱动。V_0为振幅，ω为角频率，φ_0为参考相位。基尔霍夫电压定律给出式（1.56）的回路方程，目标是求电流i(t)。

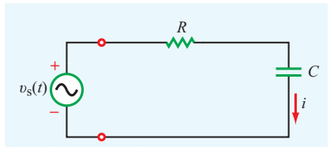


图1-20 串联RC电路；$v_s(t)$为电源电压。

$$v_s(t) = V_0 \sin(\omega t + \phi_0), \tag{1.55}$$

$$R\,i(t) + \frac{1}{C}\int i(t)\,dt = v_s(t). \tag{1.56}$$

(time domain)

式（1.56）为时域形式。直接求解包含正弦激励的积分微分方程较繁琐，下面改用相量法。

1-7.1 求解步骤

步骤1：统一采用余弦参考。为电路中各电压、电流建立共同参考，先把激励改写为余弦形式：

$$\begin{aligned} v_s(t) &= V_0 \sin(\omega t + \phi_0) \\ &= V_0 \cos\left(\frac{\pi}{2} - \omega t - \phi_0\right) \\ &= V_0 \cos\left(\omega t + \phi_0 - \frac{\pi}{2}\right), \end{aligned} \tag{1.57}$$

这里使用sin x = cos(π/2 − x)及cos(−x) = cos x。

步骤2：将时变量表示为相量。任意单频余弦函数z(t)可写为：

$$z(t) = \mathfrak{Re}\left[\widetilde{Z}\, e^{j\omega t}\right], \tag{1.58}$$

式（1.58）中Re表示取实部，Z上方加波浪号表示z(t)的相量，该相量不随时间变化。瞬时量与相量采用不同记号。
（接下页）

over the letter representing a phasor. For the voltage $v_s(t)$ given by Eq. (1.57), its phasor equivalent $\widetilde{V}_s$ is given by

$$\widetilde{V}_s = V_0 e^{j(\phi_0 - \pi/2)}. \tag{1.59}$$

Thus, $\widetilde{V}_s$ has the same amplitude as $v_s(t)$, and its exponential function is $j(\phi_0 - \pi/2)$, where $(\phi_0 - \pi/2)$ is the phase part of the cosine function in Eq. (1.57) without the ωt term. To verify that the expression for $\widetilde{V}_s$ given by Eq. (1.59) is indeed the phasor equivalent of $v_s(t)$, we apply the right-hand side of Eq. (1.58):

$$\begin{aligned}\Re\mathfrak{e}[\widetilde{V}_s e^{j\omega t}] &= \Re\mathfrak{e}[V_0 e^{j(\phi_0 - \pi/2)} \cdot e^{j\omega t}] \\ &= V_0 \cos(\omega t + \phi_0 - \pi/2),\end{aligned}$$

which is, indeed, the expression for $v_s(t)$ given by Eq. (1.57).

The time-domain and phasor-domain correspondence can be expressed in the general form

$$v_s(t) = V_0 \cos(\omega t + \phi) \quad \longleftrightarrow \quad \widetilde{V}_s = V_0 e^{j\phi}, \tag{1.60}$$

for any phase angle ϕ. Hence, ϕ may be a constant, such as $\phi_0 - \pi/2$, or it may be a function of one or more spatial variables, such as $\phi = \beta x$.

The phasor $\widetilde{V}_s$, corresponding to the time function $v_s(t)$, contains amplitude and phase information but it is independent of the time variable t. Next we define the unknown variable $i(t)$ in terms of a phasor equivalent $\widetilde{I}$ as

$$i(t) = \Re\mathfrak{e}(\widetilde{I} e^{j\omega t}). \tag{1.61}$$

If the equation we are trying to solve contains derivatives or integrals, we use the following two properties:

$$\frac{di}{dt} = \frac{d}{dt}\left[\Re\mathfrak{e}(\widetilde{I} e^{j\omega t})\right] = \Re\mathfrak{e}\left[\frac{d}{dt}(\widetilde{I} e^{j\omega t})\right] = \Re\mathfrak{e}[j\omega \widetilde{I} e^{j\omega t}], \tag{1.62}$$

and

$$\begin{aligned}\int i\, dt &= \int \Re\mathfrak{e}(\widetilde{I} e^{j\omega t})\, dt \\ &= \Re\mathfrak{e}\left(\int \widetilde{I} e^{j\omega t}\, dt\right) = \Re\mathfrak{e}\left(\frac{\widetilde{I}}{j\omega} e^{j\omega t}\right).\end{aligned} \tag{1.63}$$

► Thus, differentiation of the time function $i(t)$ with respect to time is equivalent to multiplication of its phasor $\widetilde{I}$ by $j\omega$, and integration is equivalent to division by $j\omega$. ◄

Step 3: **Recast the differential/integral equation in phasor form**

Upon using the form of Eq. (1.58) to represent $i(t)$ and $v_s(t)$ in Eq. (1.56) in terms of their phasor equivalents, we have

$$R\,\Re\mathfrak{e}(\widetilde{I} e^{j\omega t}) + \frac{1}{C}\,\Re\mathfrak{e}\left(\frac{\widetilde{I}}{j\omega} e^{j\omega t}\right) = \Re\mathfrak{e}(\widetilde{V}_s e^{j\omega t}). \tag{1.64}$$

Combining all three terms under the same real-part ($\Re\mathfrak{e}$) operator leads to

$$\Re\mathfrak{e}\left\{\left[\left(R + \frac{1}{j\omega C}\right)\widetilde{I} - \widetilde{V}_s\right] e^{j\omega t}\right\} = 0. \tag{1.65a}$$

Had we adopted a sine reference—instead of a cosine reference—to define sinusoidal functions, the preceding treatment would have led to the result

$$\Im\mathfrak{m}\left\{\left[\left(R + \frac{1}{j\omega C}\right)\widetilde{I} - \widetilde{V}_s\right] e^{j\omega t}\right\} = 0. \tag{1.65b}$$

Since both the real and imaginary parts of the expression inside the curly brackets are zero, the expression itself must be zero. Moreover, since $e^{j\omega t} \neq 0$, it follows that

$$\widetilde{I}\left(R + \frac{1}{j\omega C}\right) = \widetilde{V}_s \quad \textbf{(phasor domain)}. \tag{1.66}$$

The time factor $e^{j\omega t}$ has disappeared because it was contained in all three terms. Equation (1.66) is the phasor-domain equivalent of Eq. (1.56).

Step 4: **Solve the phasor-domain equation**

From Eq. (1.66), the phasor current $\widetilde{I}$ is given by

$$\widetilde{I} = \frac{\widetilde{V}_s}{R + 1/(j\omega C)}. \tag{1.67}$$

Before we apply the next step, we need to convert the right-hand side of Eq. (1.67) into the form $I_0 e^{j\theta}$ with I_0 being a real quantity. Thus,

$$\begin{aligned}\widetilde{I} &= V_0 e^{j(\phi_0 - \pi/2)}\left[\frac{j\omega C}{1 + j\omega RC}\right] \\ &= V_0 e^{j(\phi_0 - \pi/2)}\left[\frac{\omega C e^{j\pi/2}}{\sqrt{1 + \omega^2 R^2 C^2}\, e^{j\phi_1}}\right] \\ &= \frac{V_0 \omega C}{\sqrt{1 + \omega^2 R^2 C^2}} e^{j(\phi_0 - \phi_1)},\end{aligned} \tag{1.68}$$

where we have used the identity $j = e^{j\pi/2}$. The phase angle associated with $(1 + j\omega RC)$ is $\phi_1 = \tan^{-1}(\omega RC)$ and lies in the first quadrant of the complex plane.

（承上页）在相量字母上方加波浪号。式（1.57）电压的相量为：

$$\widetilde{V}_s = V_0 e^{j(\phi_0 - \pi/2)}. \tag{1.59}$$

该相量的模等于电压振幅；其指数为j(φ$_0$ − π/2)，其中φ$_0$ − π/2是余弦参考相位，不含ωt。代入式（1.58）右端验证：

$$\begin{aligned}\Re\mathfrak{e}[\widetilde{V}_s e^{j\omega t}] &= \Re\mathfrak{e}[V_0 e^{j(\phi_0 - \pi/2)} \cdot e^{j\omega t}] \\ &= V_0 \cos(\omega t + \phi_0 - \pi/2),\end{aligned}$$

所得结果正是式（1.57）。对任意不随时间变化的相位φ，时域与相量域的一般对应关系为：

$$v_s(t) = V_0 \cos(\omega t + \phi) \quad \longleftrightarrow \quad \widetilde{V}_s = V_0 e^{j\phi}, \tag{1.60}$$

φ可以是常数（如φ$_0$ − π/2），也可以依赖空间坐标（如βx）。相量保留振幅和相位信息，不依赖时间t。未知电流i(t)同样用其相量表示：

$$i(t) = \Re\mathfrak{e}(\widetilde{I} e^{j\omega t}). \tag{1.61}$$

若方程含时间导数或积分，可利用下列性质：

$$\frac{di}{dt} = \frac{d}{dt}\left[\Re\mathfrak{e}(\widetilde{I} e^{j\omega t})\right] = \Re\mathfrak{e}\left[\frac{d}{dt}(\widetilde{I} e^{j\omega t})\right] = \Re\mathfrak{e}[j\omega \widetilde{I} e^{j\omega t}], \tag{1.62}$$

以及

$$\begin{aligned}\int i\, dt &= \int \Re\mathfrak{e}(\widetilde{I} e^{j\omega t})\, dt \\ &= \Re\mathfrak{e}\left(\int \widetilde{I} e^{j\omega t}\, dt\right) = \Re\mathfrak{e}\left(\frac{\widetilde{I}}{j\omega} e^{j\omega t}\right).\end{aligned} \tag{1.63}$$

因此，对时间微分对应于相量乘jω；对时间积分对应于相量除以jω。校注：这里ω ≠ 0，积分式只表示同频正弦分量，积分常数与初始条件须另行处理。

步骤3：将微分 / 积分方程改写为相量方程

把式（1.56）的电流、电压代入式（1.58）形式，得：

$$R\,\Re\mathfrak{e}(\widetilde{I} e^{j\omega t}) + \frac{1}{C}\,\Re\mathfrak{e}\left(\frac{\widetilde{I}}{j\omega} e^{j\omega t}\right) = \Re\mathfrak{e}(\widetilde{V}_s e^{j\omega t}). \tag{1.64}$$

把三项合并在同一取实部算子Re之内，得：

$$\Re\mathfrak{e}\left\{\left[\left(R + \frac{1}{j\omega C}\right)\widetilde{I} - \widetilde{V}_s\right] e^{j\omega t}\right\} = 0. \tag{1.65a}$$

若采用正弦参考，并据此重新定义相量，可写成取虚部Im的形式：

$$\Im\mathfrak{m}\left\{\left[\left(R + \frac{1}{j\omega C}\right)\widetilde{I} - \widetilde{V}_s\right] e^{j\omega t}\right\} = 0. \tag{1.65b}$$

严格地说，式（1.65a）对所有t成立，故方括号内的复系数为零（ω ≠ 0），于是：

$$\widetilde{I}\left(R + \frac{1}{j\omega C}\right) = \widetilde{V}_s \quad \textbf{(phasor domain)}. \tag{1.66}$$

各项具有相同时间因子e$^{j\omega t}$，因此可将其消去。式（1.66）就是时域式（1.56）对应的相量域方程。

步骤4：求解相量域方程

由式（1.66）解得电流相量：

$$\widetilde{I} = \frac{\widetilde{V}_s}{R + 1/(j\omega C)}. \tag{1.67}$$

下一步先把结果写成I$_0$e$^{j\theta}$形式，其中I$_0$为非负实数：

$$\begin{aligned}\widetilde{I} &= V_0 e^{j(\phi_0 - \pi/2)}\left[\frac{j\omega C}{1 + j\omega RC}\right] \\ &= V_0 e^{j(\phi_0 - \pi/2)}\left[\frac{\omega C e^{j\pi/2}}{\sqrt{1 + \omega^2 R^2 C^2}\, e^{j\phi_1}}\right] \\ &= \frac{V_0 \omega C}{\sqrt{1 + \omega^2 R^2 C^2}} e^{j(\phi_0 - \phi_1)},\end{aligned} \tag{1.68}$$

这里使用j = e$^{j\pi/2}$。对于R、C、ω > 0，1 + jωRC在第一象限，其辐角φ$_1$ = arctan(ωRC)。

推导校注：不能把采用不同参考的两套相量当作同一组，从而断言实部、虚部同时为零。令方括号内系数为a + jb，式（1.65a）即a cosωt − b sinωt = 0对所有t成立，因此a = b = 0。

Phasor Technique Solution Procedure

(1) *Adopt a cosine reference* for all sinusoidally varying quantities, as in

$$x(t) = A\cos(\omega t + \phi_0).$$

- Coefficient A should be positive. If not, retain its positive magnitude and add or subtract π from ϕ_0.
- If the time function is a sine instead of a cosine, convert it using Eq. (1.57).

(2) *Convert time-dependent quantities into phasor equivalents:*

$$z(t) \quad \Longleftrightarrow \quad \widetilde{Z} \qquad \textbf{(Table 1-5)}.$$

(3) *Cast equations in the phasor domain, and then solve them for the quantity of interest.*

(4) If necessary, modify the solution $\widetilde{Y}$ into the form

$$\widetilde{Y} = Be^{j\theta},$$

where B is a positive, real quantity.

(5) *Convert the phasor-domain solution to the time domain:*

$$y(t) = \Re\mathrm{e}[Be^{j\theta}\cdot e^{j\omega t}] = B\cos(\omega t + \theta)$$

Table 1-5 Time-domain sinusoidal functions $z(t)$ and their cosine-reference phasor-domain counterparts $\widetilde{Z}$, where $z(t) = \Re\mathrm{e}[\widetilde{Z}e^{j\omega t}]$.

$z(t)$		$\widetilde{Z}$
$A\cos\omega t$	$\Longleftrightarrow$	A
$A\cos(\omega t+\phi_0)$	$\Longleftrightarrow$	$Ae^{j\phi_0}$
$A\cos(\omega t+\beta x+\phi_0)$	$\Longleftrightarrow$	$Ae^{j(\beta x+\phi_0)}$
$Ae^{-\alpha x}\cos(\omega t+\beta x+\phi_0)$	$\Longleftrightarrow$	$Ae^{-\alpha x}e^{j(\beta x+\phi_0)}$
$A\sin\omega t$	$\Longleftrightarrow$	$Ae^{-j\pi/2}$
$A\sin(\omega t+\phi_0)$	$\Longleftrightarrow$	$Ae^{j(\phi_0-\pi/2)}$
$\dfrac{d}{dt}(z(t))$	$\Longleftrightarrow$	$j\omega\widetilde{Z}$
$\dfrac{d}{dt}[A\cos(\omega t+\phi_0)]$	$\Longleftrightarrow$	$j\omega Ae^{j\phi_0}$
$\int z(t)\,dt$	$\Longleftrightarrow$	$\dfrac{1}{j\omega}\widetilde{Z}$
$\int A\sin(\omega t+\phi_0)\,dt$	$\Longleftrightarrow$	$\dfrac{1}{j\omega}Ae^{j(\phi_0-\pi/2)}$

Step 5: Convert back to the time domain

To find $i(t)$, we simply apply Eq. (1.61). That is, we multiply the phasor $\widetilde{I}$ given by Eq. (1.68) by $e^{j\omega t}$ and then take the real part:

$$i(t) = \Re\mathrm{e}[\widetilde{I}e^{j\omega t}] = \Re\mathrm{e}\left[\frac{V_0\omega C}{\sqrt{1+\omega^2R^2C^2}}e^{j(\phi_0-\phi_1)}e^{j\omega t}\right]$$
$$= \frac{V_0\omega C}{\sqrt{1+\omega^2R^2C^2}}\cos(\omega t+\phi_0-\phi_1). \quad (1.69)$$

In summary, we converted all time-varying quantities into the phasor domain, solved for the phasor $\widetilde{I}$ of the desired instantaneous current $i(t)$, and then converted back to the time domain to obtain an expression for $i(t)$. **Table 1-5** provides a summary of some time-domain functions and their phasor-domain equivalents.

Example 1-4: RL Circuit

The voltage source of the circuit shown in **Fig. 1-21** is given by

$$v_s(t) = 5\sin(4\times10^4 t - 30^\circ) \quad \text{(V)}. \quad (1.70)$$

Obtain an expression for the voltage across the inductor.

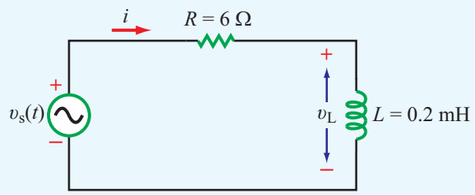


Figure 1-21 RL circuit (Example 1-4).

Solution: The voltage loop equation of the RL circuit is

$$Ri + L\frac{di}{dt} = v_s(t). \quad (1.71)$$

相量法步骤与例1-4：RL电路

相量法求解流程：

（1）统一余弦参考，写成x(t) = A cos(ωt + φ_0)。若A < 0，改用|A|并将相位增减π；正弦形式按式（1.57）转换。
（2）按表1-5把各时变量转换为相量。
（3）列出相量方程并求解所需量。
（4）把解写成模B与相位θ的极坐标形式，B为非负实数。
（5）乘回$e^{j\omega t}$并取实部，得到y(t) = B cos(ωt + θ)。

步骤5：变换回时域

按式（1.61），对式（1.68）的电流相量乘$e^{j\omega t}$并取实部，即得串联RC电路的电流：

$$i(t) = \Re\mathrm{e}[\widetilde{I}e^{j\omega t}] = \Re\mathrm{e}\left[\frac{V_0\omega C}{\sqrt{1+\omega^2R^2C^2}}e^{j(\phi_0-\phi_1)}e^{j\omega t}\right]$$
$$= \frac{V_0\omega C}{\sqrt{1+\omega^2R^2C^2}}\cos(\omega t+\phi_0-\phi_1). \quad (1.69)$$

整个过程依次为：时变量转换为相量，求解电流相量，再变换回瞬时电流。表1-5列出常见函数与余弦参考相量的对应关系；相量均用字母上方的波浪号标识。

表1-5 时域函数z(t)与相量的对应关系

$z(t)$		$\widetilde{Z}$
$A\cos\omega t$	$\Longleftrightarrow$	A
$A\cos(\omega t+\phi_0)$	$\Longleftrightarrow$	$Ae^{j\phi_0}$
$A\cos(\omega t+\beta x+\phi_0)$	$\Longleftrightarrow$	$Ae^{j(\beta x+\phi_0)}$
$Ae^{-\alpha x}\cos(\omega t+\beta x+\phi_0)$	$\Longleftrightarrow$	$Ae^{-\alpha x}e^{j(\beta x+\phi_0)}$
$A\sin\omega t$	$\Longleftrightarrow$	$Ae^{-j\pi/2}$
$A\sin(\omega t+\phi_0)$	$\Longleftrightarrow$	$Ae^{j(\phi_0-\pi/2)}$
$\dfrac{d}{dt}(z(t))$	$\Longleftrightarrow$	$j\omega\widetilde{Z}$
$\dfrac{d}{dt}[A\cos(\omega t+\phi_0)]$	$\Longleftrightarrow$	$j\omega Ae^{j\phi_0}$
$\int z(t)\,dt$	$\Longleftrightarrow$	$\dfrac{1}{j\omega}\widetilde{Z}$
$\int A\sin(\omega t+\phi_0)\,dt$	$\Longleftrightarrow$	$\dfrac{1}{j\omega}Ae^{j(\phi_0-\pi/2)}$

左列为时域函数，右列为相量；微分对应乘jω，积分对应除以jω（ω ≠ 0，积分常数另计）。

例1-4 RL电路

图1-21的电源电压如下。已知R = 6 Ω，L = 0.2 mH，求电感两端的电压表达式。

$$v_s(t) = 5\sin(4\times10^4 t - 30^\circ) \quad \text{(V)}. \quad (1.70)$$

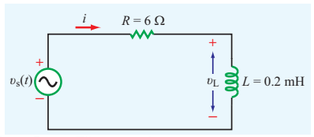


图1-21 串联RL电路，电感电压极性按原图。

$$Ri + L\frac{di}{dt} = v_s(t). \quad (1.71)$$

解：按图示电流方向与电压极性，由基尔霍夫电压定律得到式（1.71）的回路方程。（接下页）